

# SamPave V2.0 Quick Start Guide

## CRITICAL SAFETY DISCLAIMER

### IMPORTANT:

**SamPave is a professional decision-support engineering tool.** All empirical design projections, thickness suggestions, material specifications, and mechanistic calculations generated by this software are for engineering screening and decision-support only. Final pavement designs, thickness profiles, and mix configurations must be independently verified, reviewed, and signed off by a licensed and qualified Professional Pavement Engineer before field execution, bidding, or construction.

## 1. Installation and Launch (Fresh Machine)

To run SamPave, configure your local Python environment:

- 1. Run Clean Setup:** Double-click [Setup.bat](#) in the project root. This script:
  - Locates or user-installs Python 3.11+.
  - Sets up a project-local virtual environment at `.venv`.
  - Installs dependencies (PySide6, docx, openpyxl, sqlalchemy).
  - Optionally builds the standalone executable `distSamPaveSamPave.exe` via PyInstaller.
- 2. Launch the Suite:**
  - Double-click [Launch.bat](#) to run from source code.
  - Or execute the standalone binary at `distSamPaveSamPave.exe`.

## 2. 5-Minute Project Design Guide

Follow these steps to complete a flexible pavement design:

### Step 1: Start a Project

1. Click **Create New Project** on the Project Hub.
2. Enter Project Name (e.g. **National Highway NH-580**), Client (**NHAI**), and Submitted By (**QA Engineer**).
3. Select Mix Type **DBM-II** and Binder Grade **VG-30**. Click **Create Project**.

### Step 2: Input Traffic Data

1. Go to the **Traffic Analysis** tab.
2. Input:
  - CVPD = **1500**

- Growth Rate = 6.0%
- Design Life = 15 years
- VDF = 4.0
- LDF = 0.75

3. Click **Compute Traffic**. The cumulative traffic design (e.g. 38.29 MSA) will calculate instantly. Click **Save & Continue**.

### Step 3: Input Subgrade Properties

1. Go to the **Subgrade Properties** tab.
2. Input laboratory soaked CBR = 4.0%.
3. Click **Calculate Modulus**. The subgrade resilient modulus (e.g. 40.0 MPa) will calculate. Click **Save**.

### Step 4: Perform Catalogue Design

1. Go to the **Catalogue Design** tab.
2. Click **Lookup Catalogue Composition**.
3. The system queries the digitized database and populates the suggested thicknesses:
  - Bituminous Concrete (BC) = 40 mm
  - Dense Bituminous Macadam (DBM) = 160 mm
  - Wet Mix Macadam (WMM) = 250 mm
  - Granular Sub-base (GSB) = 300 mm
4. Click **Adopt Suggestion** to load these into the Structural Design panel.

### Step 5: Verify via IITPAVE & Intelligence Checker

1. Go to the **Structural Pavement Design** tab.
2. Review the thickness profile.
3. Review the **Engineering Screening Score** and warnings on the screen (powered by the Intelligence Checker).
4. If a local copy of IITPAVE.exe is configured, click **Run Mechanistic Validation** to compute strains.

### Step 6: Export DPR & Submission ZIP

1. Navigate to the **Submission Center**.
2. Change status to **Reviewed**.
3. Click **Lock Project** to lock the design parameters.
4. Click **Export DPR** to save the formatted Word report.
5. Click **Export Package (.zip)** to compile the delivery ZIP package.